

Amitrakshan Biswas

10000 58970

MAT 110 Assignment ①

Section 18

Spring 2026

①
Ques. 1

$$\lim_{x \rightarrow 2} f(x) = 5$$

As x approaches 2,

the value of $f(x)$ approaches 5

Now, for the 2nd Part of the ques. consider,

$$f(x) = \begin{cases} \frac{5}{2}x, & x \neq 2 \\ 4, & x = 2 \end{cases}$$

here $\lim_{x \rightarrow 2} f(x) = 5$ but $f(2) = 4$

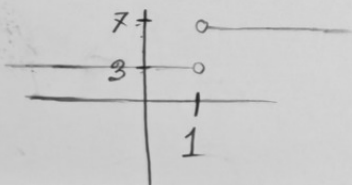
Ques. 2

$$\lim_{x \rightarrow 1^-} f(x) = 3$$

As x approaches 1 from the left
 $f(x)$ approaches 3

$$\lim_{x \rightarrow 1^+} f(x) = 7$$

As x approaches 1 from the right
 $f(x)$ approaches 7



$\lim_{x \rightarrow 1} f(x)$ does not exist.

Ques 3

a) 3

b) 1

c) Does not exist

d) $f(2) = 3$

e) 4

f) $f(4)$ not shown on graph

Ques 4

$$\begin{aligned} \text{a) } \lim_{x \rightarrow -1} \frac{x+1}{x^3+1} &= \lim_{x \rightarrow -1} \frac{\cancel{x+1}}{(\cancel{x+1})(x^2-x+1)} \\ &= \frac{1}{1+1+1} = \frac{1}{3} \quad (\text{Ans.}) \end{aligned}$$

$$\begin{aligned} \text{b) } \lim_{x \rightarrow 0} \frac{\sqrt{9+x} - 3}{x} &= \lim_{x \rightarrow 0} \frac{(9+x) - 3^2}{x(\sqrt{9+x} + 3)} \\ &= \lim_{x \rightarrow 0} \frac{\cancel{9} + \cancel{x} - 9}{x(\sqrt{9+x} + 3)} \\ &= \frac{1}{\sqrt{9+0} + 3} \\ &= \frac{1}{6} \quad (\text{Ans.}) \end{aligned}$$

Ques 4

$$\begin{aligned} \text{c) } \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} &= \lim_{h \rightarrow 0} \frac{(x^3 + 3x^2h + 3xh^2 + h^3) - x^3}{h} \\ &= \lim_{h \rightarrow 0} 3x^2 + 3xh + h^2 \\ &= 3x^2 \quad (\text{Ans.}) \end{aligned}$$

$$\begin{aligned} \text{d) } \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x} &= \lim_{x \rightarrow 0} \frac{\cancel{(1+x)} - \cancel{(1-x)}}{x(\sqrt{1+x} + \sqrt{1-x})} \\ &= \lim_{x \rightarrow 0} \frac{2x}{x(\sqrt{1+x} + \sqrt{1-x})} \\ &= \frac{2}{\sqrt{1+0} + \sqrt{1-0}} \\ &= \frac{2}{2} = 1 \quad (\text{Ans.}) \end{aligned}$$

$$\begin{aligned} \text{e) } \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{x^2+x} \right) &= \lim_{x \rightarrow 0} \frac{\cancel{(x+1)} - 1}{x(x+1)} \\ &= \lim_{x \rightarrow 0} \frac{x}{x(x+1)} = \lim_{x \rightarrow 0} \frac{1}{(x+1)} \\ &= \frac{1}{1} \quad (\text{Ans.}) \end{aligned}$$

Ques. 4

$$\begin{aligned} \text{f) } \lim_{x \rightarrow -4} \frac{\sqrt{x^2+9} - 5}{x+4} &= \lim_{x \rightarrow -4} \frac{(x^2+9) - 5^2}{(x+4)(\sqrt{x^2+9} + 5)} \\ &= \lim_{x \rightarrow -4} \frac{x^2 - 16}{(x+4)(\sqrt{x^2+9} + 5)} \\ &= \lim_{x \rightarrow -4} \frac{(x-4)(x+4)}{(x+4)(\sqrt{x^2+9} + 5)} \\ &= \frac{-8}{\sqrt{25} + 5} = \frac{-8}{10} \end{aligned}$$

$$= -\frac{4}{5} \text{ (Ans.)}$$

Ques. 5 Limits @ infinity

$\Rightarrow$ we only need to consider "highest power terms"

$$\text{a) } \lim_{x \rightarrow -\infty} \frac{(x^2+1)(2x^2-1)}{(x^2+2)} = \lim_{x \rightarrow -\infty} \frac{2x^4}{x^4} = 2$$

$$\text{b) } \lim_{x \rightarrow -\infty} \frac{3x^2 + x}{x^3 - 4x + 1} = \lim_{x \rightarrow -\infty} \frac{3x^2}{x^3} = 0$$

$$\begin{aligned} \text{c) } \lim_{x \rightarrow \infty} \frac{\sqrt{x+3x^2}}{4x-1} &= \lim_{x \rightarrow \infty} \frac{\sqrt{\frac{x}{x^2} + \frac{3x^2}{x^2}}}{\frac{4x}{x} - \frac{1}{x}} \\ &= \lim_{x \rightarrow \infty} \frac{\sqrt{\frac{1}{x} + 3}}{4 - \frac{1}{x}} = \frac{\sqrt{0+3}}{4-0} = \frac{\sqrt{3}}{4} \end{aligned}$$

$\infty - \frac{1}{\infty}$

Ques. 5

$$d) \lim_{x \rightarrow -\infty} \sqrt{4x^2 + 3x} + 2x$$

$$= \lim_{x \rightarrow -\infty} \frac{(4x^2 + 3x) - (2x)^2}{\sqrt{4x^2 + 3x} - 2x}$$

$$= \lim_{x \rightarrow -\infty} \frac{3x}{\sqrt{4x^2 + 3x} - 2x} = \lim_{x \rightarrow -\infty} \frac{\frac{3x}{|x|}}{\frac{\sqrt{4x^2 + 3x} - 2x}{|x|}}$$

$$= \lim_{x \rightarrow -\infty} \frac{\frac{3x}{-x}}{\frac{\sqrt{4x^2 + 3x}}{\sqrt{x^2}} - \frac{2x}{-x}}$$

$$|x| = \sqrt{x^2}$$

$$|x| = -x$$

when $x \rightarrow -\infty$

$$= \lim_{x \rightarrow -\infty} \frac{-3}{\sqrt{4 + \frac{3}{x}} + 2}$$

$$= \frac{-3}{\sqrt{4+0} + 2} = \frac{-3}{2+2} = -\frac{3}{4}$$

Ans.

Ques. 5

e) $\lim_{x \rightarrow \infty} \frac{\sin^2 x}{1+x^2}$ $0 \leq \sin^2(x) \leq 1$

$$\therefore 0 \leq \frac{\sin^2(x)}{1+x^2} \leq \frac{1}{1+x^2}$$

$$\lim_{x \rightarrow \infty} 0 = 0$$

$$\lim_{x \rightarrow \infty} \frac{1}{1+x^2} = 0$$

$\therefore$ Applying Squeeze Theorem

$$\lim_{x \rightarrow \infty} 0 \leq \lim_{x \rightarrow \infty} \frac{\sin^2 x}{1+x^2} \leq \lim_{x \rightarrow \infty} \frac{1}{1+x^2}$$

$$0 \leq \lim_{x \rightarrow \infty} \frac{\sin^2 x}{1+x^2} \leq 0$$

$$\therefore \lim_{x \rightarrow \infty} \frac{\sin^2 x}{1+x^2} = 0 \quad (\text{Ans.})$$

Ques. 5

$$f) \lim_{x \rightarrow \infty} (e^{-x} + 2\cos(3x))$$

$$= \lim_{x \rightarrow \infty} (e^{-x}) + \lim_{x \rightarrow \infty} 2\cos(3x)$$

$$= 0 + \underbrace{\lim_{x \rightarrow \infty} 2\cos(3x)}$$

now, this part keeps forever oscillating.

So, limit does not exist.

Ques 6

$$a) \lim_{x \rightarrow 0^-} f(x) = \cos(0) = 1 \quad \Bigg| \quad f(0) = 0$$

$$\lim_{x \rightarrow 0^-} f(x) \neq f(0) \Rightarrow \text{not continuous at } x=0$$

$$b) \lim_{x \rightarrow 1^-} f(x) = 1 - (1)^2 = 0 = f(1)$$

$$\lim_{x \rightarrow 1^+} f(x) = \ln(1) = 0 = f(1) \Rightarrow \text{yes, continuous at } x=1$$

Ques 7

a) At $x_0 = 2$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (x+2) = 4 \dots (i)$$

$$f(2) = a(2)^2 - b(2) + 3 = 4a - 2b + 3$$

to be continuous, $f(2) = (i)$

$$\Rightarrow 4a - 2b + 3 = 4$$

$$\Rightarrow 4a - 2b - 1 = 0 \dots (iii)$$

At $x = 3$

$$f(3) = 2(3) - a + b = -a + b + 6$$

$$\lim_{x \rightarrow 3^-} f(x) = a(3)^2 - b(3) + 3 = 9a - 3b + 3 \dots (ii)$$

to be continuous, $f(3) = (ii)$

$$-a + b + 6 = 9a - 3b + 3$$

$$b + 6 = 10a - 3b + 3$$

$$6 = 10a - 4b + 3$$

$$0 = 10a - 4b - 3 \dots (iv)$$

Ques 7

a) Continued...

Solving Simultaneous Equations (iii) & (iv)

we get $a = \frac{1}{2}$ $b = \frac{1}{2}$ (Ans.)

b)

$$f(x) = \frac{x^2 + 9}{x - 3}$$

is not continuous, where
the denominator is zero.

i.e. at $x = 3$

$$\lim_{x \rightarrow 3} \frac{x^2 + 9}{x - 3} = \lim_{x \rightarrow 3} \frac{(x+3)(x-3)}{\cancel{(x-3)}} = 3+3 = 6$$

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$$

$\therefore$ Discontinuity is removable!

(Ans.)

Ques 8

$$8.a) \quad y = \sin\left(2 \tan^{-1} \sqrt{\frac{1-x}{1+x}}\right) \quad \text{let } x = \cos \theta$$

$$= \sin\left(2 \tan^{-1} \sqrt{\tan^2 \frac{\theta}{2}}\right)$$

$$\frac{1-\cos \theta}{1+\cos \theta} = \tan^2\left(\frac{\theta}{2}\right)$$

$$= \sin\left(2 \cdot \frac{\theta}{2}\right)$$

$$= \sin \theta$$

$$= \sqrt{1-\cos^2 \theta}$$

$$\Rightarrow \dot{y} = (1-x^2)^{1/2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2} \cdot (1-x^2)^{-1/2} \cdot (-2x) = -\frac{x}{\sqrt{1-x^2}}$$

$$8.b) \quad y = \ln \sqrt{\frac{1+\sin x}{1-\sin x}} = \frac{1}{2} (\ln(1+\sin x) - \ln(1-\sin x))$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2} \left[\frac{1}{1+\sin(x)} \cdot (\cos x) - \frac{1}{1-\sin(x)} \cdot (-\cos x) \right]$$

$$= \frac{\cos(x)}{2} \left[\frac{1}{1+\sin(x)} + \frac{1}{1-\sin(x)} \right]$$

$$= \frac{\cos(x)}{2} \left[\frac{1-\sin(x)+1+\sin(x)}{1-\sin^2(x)} \right] = \frac{\cos(x)}{2} \cdot \frac{2}{\cos^2(x)} = \sec(x)$$

Ques. 8

$$\begin{aligned} 8.c) \quad y &= \ln\left(\frac{1}{\sqrt{\cos(x)}}\right) = \ln(1) - \ln(\sqrt{\cos x}) \\ &= 0 - \frac{1}{2} \ln(\cos x) \end{aligned}$$

$$\Rightarrow y = -\frac{1}{2} \ln(\cos x)$$

$$\frac{dy}{dx} = \left(-\frac{1}{2}\right) \cdot \frac{1}{\cos x} \cdot (-\sin x)$$

$$= \frac{1}{2} \cdot \frac{\sin x}{\cos x} = \frac{1}{2} \tan(x) \quad (\text{Ans.})$$

$$8.d) \quad x = a \sec^2 \theta$$

$$\frac{x}{a} = \sec^2 \theta$$

$$y = a \tan^2 \theta$$

$$\frac{y}{a} = \tan^2 \theta$$

$$\sec^2 \theta - \tan^2 \theta = 1 = \frac{x}{a} - \frac{y}{a} = \frac{x-y}{a}$$

$$\Rightarrow a = x-y$$

$$\Rightarrow y = x-a$$

$$\Rightarrow \frac{dy}{dx} = 1 - 0 = 1$$

(Ans.)

e)

Ques. 9

i) $f(x) = x^4 - 8x^2 + 16$

$$f'(x) = 4x^3 - 16x + 0 = 4x^2$$

$$= 4x(x^2 - 4)$$

$$= 4x(x-2)(x+2) \Rightarrow x\text{-intercepts } 0, 2, -2$$

$$\Rightarrow (-\infty, -2) \quad -ve$$

$$(-2, 0) \quad +ve$$

$$(0, 2) \quad -ve$$

$$(2, \infty) \quad +ve$$

$$f'(x) = 4x^3 - 16x$$

$$f''(x) = 12x^2 - 16$$

$$= 4(3x^2 - 4)$$

$$\text{inflection point} = \pm \frac{2}{\sqrt{3}}$$

a) increasing : $(-2, 0)$ and $(2, \infty)$

b) decreasing : $(-\infty, -2) \cup (0, 2)$

c) concave up : $(-\infty, -\frac{2}{\sqrt{3}})$ and $(\frac{2}{\sqrt{3}}, \infty)$

d) " down : $(-\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}})$

e) inflection @ : $\pm \frac{2}{\sqrt{3}}$

critical @ : $0, -2, 2$

f) Minima : $(-2, 0)$ and $(2, 0)$

Maxima : $(0, 16)$

Ques 9

$$\text{ii) } f(x) = \frac{x^2}{x^2+2}$$

$$f'(x) = \frac{(x^2+2)(2x) - (x^2)(2x)}{(x^2+2)^2} = \frac{2x(x^2+2-x^2)}{(x^2+2)^2}$$

$$\underline{\underline{x\text{-intercept}}}: 0 \quad = \frac{4x}{(x^2+2)^2}$$

Denominator is never 0

$(-\infty, 0)$ -ve

$(0, \infty)$ +ve

$$\begin{aligned} f''(x) &= \frac{(x^2+2)(4) - (4x)(x^2+2)(2)(2x)}{(x^2+2)^4} \\ &= \frac{4(x^2+2) - 16x^2}{(x^2+2)^3} \end{aligned}$$

inflection points:

$$8 - 12x^2 = 0$$

$$x^2 = \frac{-8}{-12} = \frac{2}{3}$$

$$x = \pm \sqrt{\frac{2}{3}}$$

$$= \frac{8 - 12x^2}{(x^2+2)^3}$$

a) Increasing: $(0, \infty)$

b) Decreasing: $(-\infty, 0)$

c) Concave up: $(-\sqrt{\frac{2}{3}}, +\sqrt{\frac{2}{3}})$

d) " Down: $(-\infty, -\sqrt{\frac{2}{3}}) \cup (\sqrt{\frac{2}{3}}, \infty)$

e) x-coordinates:

inflection: $\pm \sqrt{\frac{2}{3}}$

critical: 0

f) minima @ $(0, 0)$

no maxima.

Ques 9 (iii)

$$f(x) = \sqrt[3]{x+2} = (x+2)^{1/3}$$

$$f'(x) = \frac{1}{3} \cdot (x+2)^{-2/3} \cdot (1) = \frac{1}{3(x+2)^{2/3}}$$

$f'(x)$ is undefined @ Critical Points: $x = -2$

$f'(x)$ is always Positive. (Denominator is Squared)

$$f''(x) = \frac{1}{3} \cdot \left(-\frac{2}{3}\right) \cdot (x+2)^{-5/3} \cdot (1) = \frac{-2}{9(x+2)^{5/3}}$$

Inflection Point $x = -2$

$f''(x)$ is -ve $x < -2$

$f''(x)$ is +ve $x > -2$

- a) increasing: $(-\infty, \infty)$
- b) decreasing: from left to right, never decreasing.
- c) concave up: $(-\infty, -2)$
- d) down: $(-2, \infty)$
- e) inflection: $x = -2$ Critical: $x = -2$
- f) no maxima/minima.

Ques 10

a) $f(x) = x^2 - 6x + 8$ on $[2, 4]$

① Continuity: yes, $f(x)$ is polynomial

② Differentiability: yes, on open-interval $(2, 4)$

③ Endpoints : $f(2) = 4 - 12 + 8 = 0$

$$f(4) = 16 - 24 + 8 = 0$$

$$\therefore f(2) = f(4)$$

$$\therefore f'(x) = 2x - 6 = 0 \quad @ \quad x = 3$$

Rolle's Theorem works! Theorem works!

b) $f(x) = x^3 + x - 4$ on $[-1, 2]$

$$f'(x) = 3x^2 + 1$$

① Continuity: yes, $f(x)$ is polynomial

② Differentiability: yes, on $(-1, 2)$

$$f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{f(2) - f(-1)}{2 - (-1)} = \frac{6 - (-6)}{3}$$

$$f(2) = 8 + 2 - 4 = 6$$

$$f(-1) = -1 + (-1) - 4 = -6$$

$$= \frac{12}{3} = 4$$

$$c = 1$$